

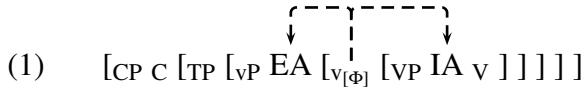
Person hierarchy effects from Φ -agreement at the left periphery in Kawahíva*

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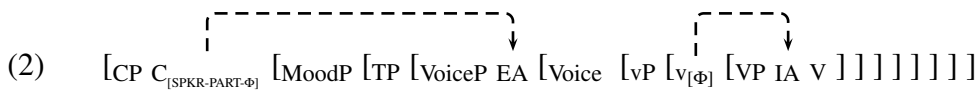
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1. Introduction

In general, person hierarchy effects result from a probe agreeing with multiple goals, typically in the low domain of the clause (Béjar 2003, Béjar and Rezac 2009:a.m.o), as schematized in (1).



However, in the Tupí-Guaraní language Kawahíva, spoken in the Brazilian Amazon, the person hierarchy $1>2>3$ manifests only when the verb moves to CP, overtly or covertly. Otherwise, agreement is restricted to the internal argument. To account for this pattern, I propose Kawahíva matrix clauses come with two probes along the clause spine, as displayed in (2): a probe relativized to [SPKR-PART- Φ] sits on C, and a flat Φ -probe, is on v .



This analysis is used to explain i) the lack of person hierarchy effects when the verb does not move to the CP in matrix clauses; ii) the absence of subject agreement and person hierarchy effects in dependent clauses; and iii) an asymmetry between subject and object agreement markers in verbal reduplication. If on the right track, this proposal extends other accounts for multiple loci for Φ -agreement in the clausal spine (Coon 2013, Oxford 2015, Compton 2018).

The paper proceeds as follows: After presenting basic information about the language and the data provenance in section 2.1, I provide an overview of the clause structure in

*This work would not have been possible without Mandeí Juma and Awip (Davi) Uru Eu Wau Wau. I am also thankful to Line Mikkelsen and Peter Jenks for their valuable feedback on the project that led to this work.

2.2, and about the relevant agreement data, in 2.3. In 3, I lay out my proposal to capture the person hierarchy effects in Kawahíva. Crucially, this account posits that the relativized probe that leads to these effects resides on C, not *v*. Then, in sections 3.1 through 3.3, I discuss how this proposal accounts for several facts that would not have a principled explanation in the alternative account.

2. Kawahíva basics

2.1 Language and data provenance

Kawahíva (or Kagwahíva, in varieties with the phoneme /g^w/) is a highly endangered Tupian language (in the Tupí-Guaraní branch) spoken by some 560 people within a population of about 1060 people in the Brazilian Amazon (in the states of Amazonas, Mato Grosso, and Rondônia) (dos Santos 2024a). There are currently eight ethnolects of the language, all named after the communities (ISO codes are provided within brackets when available): Amondawa [adw]; Jiahui, Parintintin, Tenharin [pah]; Juma [jua]; Jupaú [urz]; Karipuna [kuq]; and Piripkura.

Some of the previous works on Kawahíva grammar, including those on different varieties, and unrelated to the present topic, include Pease (1968) and dos Santos (2022, 2023, 2024c,b, 2025).

Data for this paper come from texts and follow-up elicitation on two varieties, Júma and Jupaú (also known as Uru Eu Wau Wau), collected by the author during in situ fieldwork with several speakers, from 2019 to the present. Recordings of work sessions are archived at the California Language Archive.¹

2.2 Clause structure

Pragmatically-neutral matrix clauses (e.g., event reports and out-of-the-blue clauses) exhibit a VSO order in Kawahíva, as shown by the event report example in (3).² Other word orders are possible following the fronting of other phrases in the sentence. For instance, OSV is the result of object extraction in, e.g., questioning or (exhaustive) focus, as illustrated in (4-5). Furthermore, the verb and fronted phrases compete for the initial slot – when initial, only one or the other can immediately precede a mood particle (*po* in 3) and a tense particle (*txi* in 3), in this order, if both particles are present.

¹The collection is available online: <http://dx.doi.org/doi:10.7297/X2P26W9H>.

²Glossing conventions follow the Leipzig Glossing Rules, with the following additions: SET A ‘subject agreement’; SET B ‘object agreement’; IRR ‘irrealis mood’; N = ‘nominal’; NOM.FUT = ‘nominal future’; NOM.NEG = ‘nominal negation’; NOM.PST = ‘nominal past’; NWIT = ‘non-witnessed’; PROSP = ‘prospective’; PRO ‘pronoun’; PROX ‘proximal’; REAL ‘realis mood’; RED ‘reduplicant’; REM = ‘remote’. Examples are presented using the Kawahíva orthography, which is similar to the IPA with the following exceptions: ⟨y⟩ = [i], ⟨ʻ⟩ = [ʔ], ⟨kw⟩ [k^w], ⟨tx⟩ = [tʃ], ⟨ngw⟩ = [ŋg^w], ⟨gã⟩ = [ŋã].

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- (3) V-epiang po txi ji=rembiriko=ga Kaninde=gã nãhẽ.
 3.A-see IRR FUT 1.SG.PRO=partner=3.SG.MASC Kanindé.NGO=3.PL PROSP
 ‘My husband might see the (people from the NGO) Kanindé.’
- (4) Mãngã po ji=ruvyra v-epiang rimba’e?
 who IRR 1.SG.POSS=uncle 3.A-see DISTANT.PST
 ‘Who did my uncle see back in the olden times?’
- (5) Pirapetxinguhua te’i ki hẽa o-pyhy-pyhy ko.
pirapetxinguhua.fish only PST 3.SG.FEM.PRO 3.A-catch-RED REAL
 ‘It was only *pirapetxinguhua* fish that she caught.’

Assuming a layered approach to clause structure, the matrix clause can be represented as in (6). The CP, MoodP, and TP projections are supported by the properties of matrix clauses mentioned above, namely the single slot in clause-initial position before the exponents of mood and tense, and the particles and their rigid order following this initial position. The other projections, VoiceP, ν P and VP, are unexceptional on the layered approach to clausal structure arrived at in various schools of syntactic analysis (Kratzer 1996, Pyllkkänen 2002, Harley 2013).

- (6) CP – MOODP – TP – VoiceP – ν P – VP

Dependent clauses exhibit several structural differences when compared to matrix clauses in Kawahíva. For instance, they exhibit a strict SOV order, as illustrated by the complement clause within brackets in (7); this is also true for relative clauses and adjunct clauses, not shown here.

- (7) E-mõmbe’u [tapy’ynha-gã pira i-pyhyk-aver=a ore=kavyra pyri]!
 1.SG.A-tell non-indigenous fish i-catch-NOM.PST=N 1.EXCL=land near
 ‘Tell (them) that the non-indigenous people caught fish in our land!’

Other structural differences include the absence of properties related to the top projections in matrix clauses, CP and MoodP. These properties are the absence of verb and phrasal fronting and the absence of the exponents of MoodP. In the next section, we will also see that these clauses do not exhibit subject agreement, which is associated with CP. I propose that these facts are consistent with the structure as represented in (8), where dependent clauses are structurally reduced from full clauses.

- (8) TP – VoiceP – ν P – VP

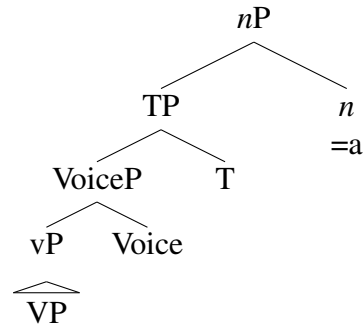
Dependent clauses are mixed nominalizations, involving a mix of nominal and verbal properties (akin to English *-ing* gerunds) (Alexiadou 2020, Iordăchioaia 2020:among many others). Beginning with their verbal properties, these nominalizations show adverbial mod-

ification in the form of verbal suffixes (9-10) and locative postpositional phrases (11), voice morphology in the form of causative (12), reflexive (13), and reciprocal (14), pluractional aspect in the form of reduplication (14), negation (15), and tense morphology in the form of the temporal past suffix *-(a)ver* (9-15). Additionally, they all have event participants (subjects and objects) and event readings. Turning to the nominal properties of these structures, they occur in the object argument position. Another diagnostic is language-internal: the tense and negation morphology is context-sensitive – they are nominal-sensitive allomorphs, conditioned by features of the nominal-building morpheme *=a*.

- (9) E-mõmbe'u [tapy'ynha=gã rur-**ahy**-aver=*a*]!
 2.SG.IMP-tell non-indigenous=3.PL come-forcefully-NOM.PST=N
 'Tell (them) that the non-indigenous people came (on a motorcycle) crazily.'
- (10) E-mõmbe'u [gã mãjioka rerur-**atehe**-ver=*a*]!
 2.SG.A-tell 3.PL.PRO manioc bring-IN.VAIN-NOM.PST=N
 'Tell (them) they brought manioc for no reason.'
- (11) E-mõmbe'u [tapy'ynha=gã pira i-pyhyk-aver=*a* **ore=kavyra pyri**]!
 2.SG.A-tell non-indigenous=3.PL fish i-catch-NOM.PST=N 1.EXCL=land near
 'Tell (them) that the non-indigenous people caught fish in our land!'
- (12) E-mõmbe'u [ga mbahira i-**mbo**-vev-aver=*a*]!
 2.SG.IMP-tell 3.SG.MASC.PRO light i-CAUS-turn.off-NOM.PST=N
 'Tell (them) that he turned off the light!'
- (13) E-mõmbe'u [gã **j**-apo-aver=*a*]!
 2.SG.IMP-tell 3.PL.PRO REFL-do-NOM.PST=N
 'Tell (them) that they hurt themselves!'
- (14) E-mõmbe'u [gã **jo**-mõngita-mõngita-ver=*a*]!
 2.SG.IMP-tell 3.PL.PRO RECP-narrate.story-RED-NOM.PST=N
 'Tell (them) that they repeatedly narrated stories to each other!'
- (15) E-mõmbe'u [gã mãjioka rerur-**e'ym**-aver=*a*]!
 2.SG.IMP-tell 3.PL.PRO manioc bring-NOM.NEG-NOM.PST=N
 'Tell (them) that they did not bring manioc!'

Additionally, I assume the clause-embedding nominalizer is *=a*. This morpheme is regarded as a nominal-exclusive morpheme in all approaches to the cognate morpheme in the Tupí-Guaraní language family, sometimes even being considered as a noun-forming morpheme (Rodrigues 1953, Lemos Barbosa 1956, Rodrigues 1996, 2001, Vieira 1995, Queixalós 2006). After we update the structural representation for dependent clauses to accommodate these facts about them, we arrive at the diagram in (16).

(16)



2.3 Agreement

In the Juma and Jupaú ethnolects of Kawahíva, matrix verbs generally index either the subject (17) or object (18); the choice depends on a person hierarchy $1 > 2 > 3$ (dos Santos 2023). The complete set of subject markers (SET A), which are inflectional prefixes, and that of object markers (SET B), which are bound pronouns (SET B), is provided in Table (19). Note that Kawahíva verbs agree with 3rd person subjects, but not 3rd person objects.

- (17) A-hepia ki ji pe ko.
1.SG.A-see PST 1.SG.PRO 2.PL.PRO REAL
'I saw you (PL.).'

- (18) Ji=repia ki pe ko.
1.SG.B=see PST 2.PL REAL
'You (PL.) saw me.'

- (19) *Kawahíva agreement markers*

	SET A	SET B
1ST PERSON SG.	a-	ji=
2ND PERSON SG.	ere-	nde=
1ST PERSON INCL.	txi-	nhãnde=
1ST PERSON EXCL.	oro-	ore=
2ND PERSON PL.	pe-	pe=
3RD PERSON	v- ~ o-	–

3. An account of the person hierarchy in Kawahíva

I argue that the person hierarchy effects arise from a higher probe on C, not v , contrary to accounts of hierarchy effects where the relevant probe is on v . In particular, the probe on C is relativized to [SPKR-PART- Φ]; the probe on v , however, is [Φ]. Assuming the matrix clause structure proposed in 2.3 above, and the proposed loci for the probes, we arrive at the structure exhibited in (20).

- (20) [CP C_[SPKR-PART-Φ] [MoodP [TP [VoiceP EA [Voice [vP [_{v[Φ]}] [VP IA v]]]]]]]]

This proposal predicts that, if the verb does not raise to CP, it will not exhibit subject agreement. We will see that this is true. There is one instance where matrix verbs cannot agree with the subject, and the person hierarchy is neutralized. These are cases where the clause-initial position is filled with an adverbial PP. We will also see that this account explains a cluster of other properties regarding verbal agreement.

3.1 Verb movement to CP

The appearance of person hierarchy effects is dependent on verb movement to CP, overtly or covertly.

dos Santos (2025) has established that Kawahíva exhibits long distance verb movement to CP in VSO matrix clauses and clauses with DP extraction. This movement follows from a categorial feature [V] with the EPP property on C. This is schematized in (21).

- (21)
$$[_{CP} C \underset{\substack{[VP \text{ PART-}\Phi] \\ (\bar{A})}}{[V]} [_{MoodP} [_{TP} [_{VoiceP} EA [_{Voice} [_{vP} [_{v[\Phi]} [_{VP} IA V] }]]]]]]]]$$

In VSO matrix clauses, the verb moves overtly to CP. In DP extraction contexts, the verb moves to CP, but it is spelled out in an intermediate lower position, as it precedes the internal argument. That is, the verb moves covertly in contexts of DP extraction. Furthermore, in VSO matrix clauses and instances of DP extraction, the verb shows agreement with the subject, and person hierarchy effects are present, as in (22).

- (22) Mãngã po v-epia gã rimba'e?
 who IRR 3.A-see 3.PL NWIT.REM.PST
 'Who saw them back in the day?'

However, in contexts of adverbial PP (non-DP) extraction, the verb does not move at all. It appears in a lower position, but unlike the DP extraction scenarios, it follows the internal argument and cannot bear subject agreement. Crucially, person hierarchy effects disappear, and agreement is restricted to the object, as (23) shows. A summary of the differences correlated with the type of extractee is shown in (24).

- (23) Oi'i=ve ki jie gã (*a)-repiak-i.
 some.time.ago=to PST 1.SG.PRO 3.PL.PRO (*1.SG.A)-see-i
 'The other day, I saw them.'

These facts, including the lack of subject agreement, follow naturally under the proposal that dependent clauses are truncated, projecting up to TP, as schematized in (28). Consequently, dependent clauses do not come with the higher probe that leads to the person hierarchy effects, the CP. On the alternative proposal that person hierarchy effects derive from a lower probe, these facts would not receive a principled explanation.

(28) [TP [VoiceP EA [Voice [vP [v_[\Phi]] [VP IA V]]]]]]

3.3 Application to verbal reduplication

A further advantage of this proposal concerns an asymmetry involving the person markers and verbal reduplication. In Kawahíva, object markers, but not subject markers, must be copied into the reduplicant (dos Santos 2023). We can see this pattern in examples (29-30).

(29) Are-rety-[areretyk] ki gã ko.
1.EXCL.B=pull.down-[RED] PST 3.PL REAL
'They pulled us all down.'

(30) *Are=rety-[retyk] ki gã ko.
1.EXCL.B=pull.down-[RED] PST 3.PL REAL
'They pulled us all down.'

In contrast, subject agreement markers do not have a similar requirement, as shown in (31-32).

(31) Anhãnga o-kutu-[kutu]-pam j-urua.
ghost 3.A-pierce-[RED]-COMPLETELY i-mouth
'It was the ghost that pierced his mouth (the chief's mouth) multiple times'

(32) *Anhãnga o-kutu-[okutu]-pam j-urua.
ghost 3.A-pierce-[RED]-COMPLETELY i-mouth
'It was the ghost that pierced his mouth (the chief's mouth) multiple times'

In the current proposal for a relativized probe on C, this fact receives a straightforward account: reduplication targets v , which does not include the probe on C.

4. Remarks

Positing two distinct Φ -probes in Kawahíva matrix clauses – one probe relativized to [SPKR-PART- Φ] on C and one flat probe on v – provides a unified account for several disparate phenomena in the language, including person hierarchy effects only when the verb moves to C in matrix clauses, the total lack of subject agreement and the said effects in dependent clauses, and the requirement that reduplication copies object agreement, but not subject agreement. Additionally, this work extends the typology of where Φ -probes can occur in the clausal spine.

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